M269: Algorithms, Data Structures and Computability

Tutor-Marked Assignment (TMA)
Fall 25/26

Cut-Off Date: Based on the Published Deadline.

Total Marks: 60 marks turned to 10 marks

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Plagiarism Warning:

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I hereby declare that this submitted TMA work is a result of my own efforts and I have not plagiarized any other person's work. I have provided all references of information that I have used and quoted in my TMA work.

Name of Student:……..

Signature:……...

Date:……

Question 1: (30 marks)

In a parking area of certain restaurant, the arrival and departure times of customers' cars are recorded. Using this data, it is requested to find the minimum number of parking spots that should be there to avoid losing prospective customers who may leave because they cannot find place to park their cars.

Examples:
Ex1:
Inputs:
Arrivals = [9:00, 10:00, 11:00, 12:00, 14:00, 19:00]
Departures = [9:50, 15:00, 15:50, 16:50, 16:00, 20:00]
Output:
The minimum number of parking slots is 4.

Ex2:
Inputs:
Arrivals = [700, 950, 1270, 1400, 1500, 2000]
Departures = [900, 1250, 1300, 1800, 1700, 2100]
Output:
The minimum number of parking slots is 2.

a) Design an algorithm to solve this problem **(10 marks)**,
b) Analyse its complexity of your algorithm **(4 marks)**
c) Develop a python code to implement your efficient algorithm. (**10 marks**) [The marks depend on the correctness of the code, indentation, comments, test-case]
d) Prepare a brief report (250 words) comparing your algorithm with a brute force one **(6 marks)**

Question 2: (30 marks)

There are some people standing in a queue You are given their heights in an array A[]. For each person in the queue, it is required to search for the next tallest person in the queue and to print the difference of height their heights. Print all the outputs in an array Out[ ].

Note:

- The last one in the queue is then compared with the first in the queue as if it is a circular queue.
- Print -1 if no one is taller than the person under consideration.

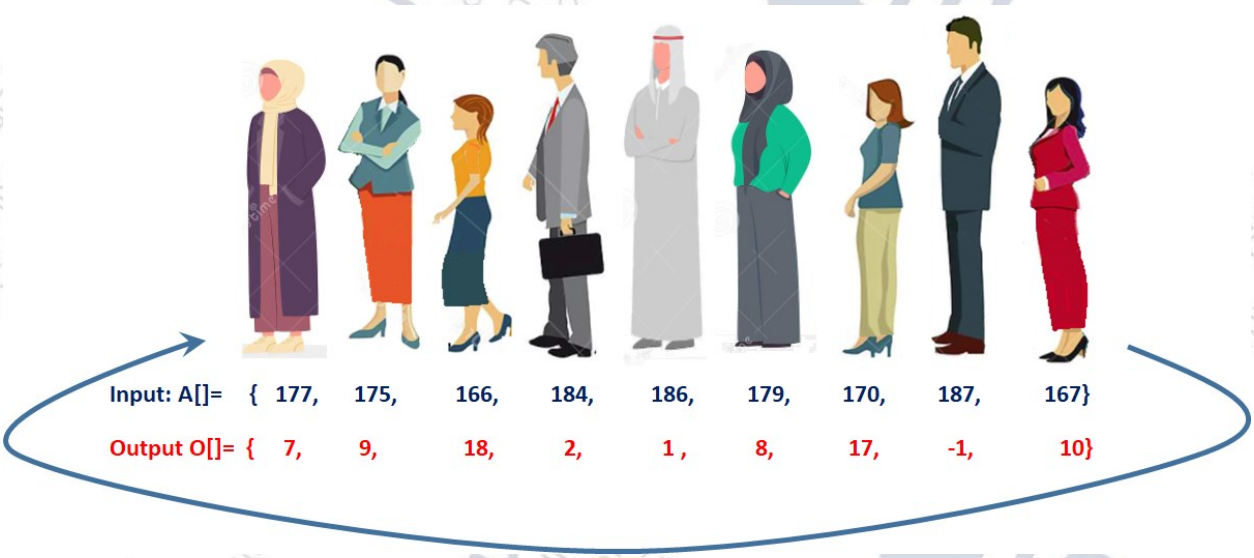
Example 1:
(refer to the figure below)
A[ ]={177, 175, 166, 184, 186, 179, 170,187,167}
Output:
Out[ ]={7, 9, 18, 2, 1, 8, 17, -1, 10}

Example 2:
A[]={174,180,184,160,177}
Output:
Out[ ]={6, 4, -1, 17, 3}

Explanation of example 1:
The given array A[] of heights is as follows:

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]
177	175	166	184	186	179	170	187	167

The lady at the starting point of the queue has a height 177 (i.e., A[0]). The next first person in the queue taller than her is 184 (i.e., A[3]). The difference between their heights is 184 - 177=7, Therefore 7 is printed. The second one with height 175 (i.e., A[1]) has also the same person as the next first taller person. Therefore 9 is printed and so on. Remember that the queue is circular. Therefore, the last lady in the queue with height 167 (i.e., A[8]) is compared with the people at the starting points of the queue, where the lady with height 177 (i.e., A[0]) is the next first taller person, and hence, 10 is printed.



a) Design a more efficient algorithm to solve this problem. (**10 marks**) [The efficiency of your algorithm is the main driver of the mark],
b) Analyze the complexity of your solution. (**4 marks**) [full explanation of your answer should be provided]
c) Develop a python code to implement your efficient algorithm. (**10 marks**) [The marks depend on the correctness of the code, indentation, comments, test-case]
d) Prepare a brief report (250 words) comparing the you algorithm with a brute force one (**6 marks**)

Note:
You should submit all your answers for the above-mentioned questions in **one big word file** (one after the other). The following table shows how your answers are evaluated:

Criterion	Unsatisfactory	Average	Good
Algorithm Design	Incomplete.	Complete but looks like an essay not a professional algorithm.	100% Complete and in a professional format Explanation of the pseudo-code is provided looks efficient in terms of running time and of memory usage Comments on the efficiency
Implementation Using Python	Incomplete Poor use of white space (indentation, blank lines, ambiguous naming of variables)	Organized work, with correct use of space, variables, and naming.	Completed 100% Comments on the code Screen shot of the run and comments. Organized indentation with good use of variables Tested with any simple input case study. Documented briefly in terms of (input, output, functions purposes)
Complexity analysis	No answer	Incomplete answer or partially correct.	Completed 100%. Explanation is provided.
Report	Irrelevant text or no answer.	some effort but the report is not organized	Logical representation of the arguments in terms of problem's categories, solving technique, complexity, memory consumption and ending with a correct conclusion.